MATH0104 Modular Forms

Year:	2026–2027
Code:	MATH0104
Level:	7 (UG), 7 (PG)
Normal student group(s):	UG: Y3, Y4 Mathematics programmes PG: MSc Mathematics
Value:	15 credits (7.5 ECTS credits)
Term:	1
Assessment:	70% examination, 30% departmental test(s)
Normal Pre-requisites:	MATH0013 (Analysis 3: Complex Analysis) MATH0014 (Algebra 3: Further Linear Algebra)
Lecturer:	Dr O Gregory

Course Description and Objectives

This module aims to offer an overview of the basic notions that appear in the classical theory of modular forms. These are analytic objects encoding a lot of arithmetic information which makes them a central point of study in number theory and arithmetic geometry. Modular forms also arise naturally in a variety of other research fields like transcendence proofs, differential equations and mirror symmetry.

The main objects of interest are functions on the complex upper half plane that transform in a special way under the action of $SL_2(\mathbb{Z})$. Two concrete constructions of such functions will be covered: Eisenstein series and theta series. We will also show that the space of modular forms for a specific weight is finite dimensional, which makes all these functions algorithmically computable. Furthermore, we will introduce Hecke operators and show that the L-functions associated to eigen functions are of arithmetic nature. Throughout the module, several applications to congruences and positive definite quadratic forms will be highlighted.

Recommended Texts

– LJP Kilford, *Modular Forms: A Classical and Computational Introduction* (Imperial)

– JP Serre, *A Course in Arithmetic* (Springer)

– W Stein, *Modular Forms, a Computational Approach* (AMS)

– H Cohen and F Stromberg, *Modular Forms: A Classical Approach* (AMS)

Detailed Syllabus

– The modular group

– Modular functions and modular forms for the full modular group

– The dimension of the space of modular forms of weight k

– Hecke operators

– The Petersson Inner Product.

– The L-function of a modular form

- Theta functions
- Applications to quadratic forms

September 2026